A quick review of cryptography fundamentals will help you understand the private-public key pair that is used in addressing the problem of decentralized identities for unknown participants. Unlike in a traditional system, in a decentralized system, you cannot use a username-and-password approach to identify and authenticate a user. Instead, the technique that is often used is similar to accessing a server instance on a cloud provider, using cryptographic key pairs.

5.1.1 Symmetric key cryptography

Let's start by taking a quick look at symmetric key cryptography so you understand the encryption process and also why this method may not be suitable for decentralized applications. It's called *symmetric key* encryption because the same key is used for encryption and decryption. Let's examine the common Caesar encryption. In this encryption, the individual letters of a message are alphabetically shifted by a fixed number (key). Consider the message in figure 5.2: `Meet me at the cinema`. You shift every letter by 3 to encrypt it; the receiver of your message decrypts it by using the same "key" and shifting each letter the other way by 3 to view the original message.

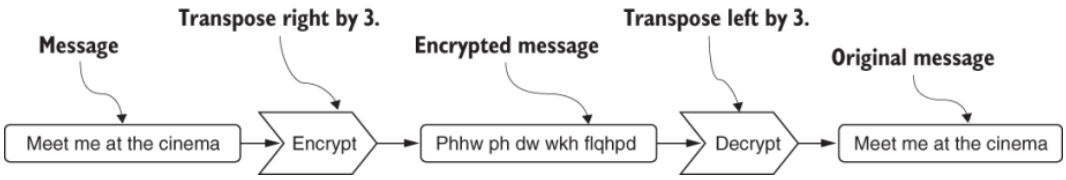


Figure 5.2 Symmetric key encryption

In this trivial example, 3 is the encryption key. Because the same key is used for encryption and decryption, it's symmetric key encryption. The key and the encryption and decryption functions are typically much more complex in a real application. Regardless, symmetric encryption has a significant issue: that of key distribution, or how to pass the key to the participants secretly. If you make it public, then anybody can decrypt the message. This issue is further exacerbated in a blockchain-based decentralized network, in which you're dealing with unknown